

ATLAS: Anytime-valid Testing of Latent Simulators

Sequential Certification of World-Model Faithfulness via E-Process Betting

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Abstract

A world model—a learned, action-conditioned simulator of an environment—is useful only insofar as its imagined rollouts remain *faithful* to reality. Yet faithfulness is a moving target. It *decays with the rollout horizon*, because recursive prediction compounds one-step error, and it *drifts over deployment time*, because of non-stationarity, the sim-to-real gap, and the self-referential feedback loop of model-based reinforcement learning, in which an improving policy continually pushes the agent into regions the model never learned. The prevailing practice—scoring a world model once, offline, with metrics such as Fréchet Video Distance, PSNR, or held-out return gaps—is structurally incapable of tracking this degradation: an offline number computed before deployment says nothing about whether the model is still trustworthy a thousand steps later, at horizon twenty, after the policy has changed. We introduce **ATLAS**, a framework that equips a deployed world model with a *running, statistically valid* certificate of the form “as of now, this model’s rollouts are trustworthy up to horizon $h^*(t)$ at tolerance ε ,” with a guarantee that holds *uniformly over all time*. ATLAS scores each horizon with a strictly proper scoring rule—which makes it agnostic to the model’s representation, covering pixel, latent, and embedding predictors alike—and tests the null hypothesis that the horizon is ε -faithful by betting against it with a test martingale. Combining the resulting per-horizon e-processes yields the *faithfulness frontier* $h^*(t)$: a time-evolving trust horizon that a planner may legally use to clip its imagination depth, turning a passive monitor into an active controller. Crucially, the construction is *conditional*—it places one bet at a time against a conditional-mean null—so it never invokes independence or exchangeability and remains valid on the endogenous, policy-dependent data stream that breaks conformal methods. We establish three guarantees: (i) a Ville-type theorem that a genuinely faithful horizon is revoked with probability at most α , uniformly over all rounds; (ii) an *anytime-valid simulation lemma* that converts the running, empirically certified one-step error into a time-uniform bound on the h -step value gap, forging the missing link between sequential testing and model-based RL theory; and (iii) an order-optimal bound on the delay to detect a faithfulness breach. We prove all three in full. Empirically, we first validate each theorem on a controllable latent simulator where ground truth is known; we then demonstrate ATLAS on real image sequences (Moving MNIST and the KTH human-action dataset) under three qualitatively different distribution shifts; and finally we monitor *learned neural* world models—a convolutional autoencoder and a JEPA-style frozen-encoder model—trained on a GPU. In every setting the trust horizon collapses precisely when reality diverges from imagination, while offline metrics remain blind.

Contents

1	Introduction	2
2	Related work	4
3	Problem formulation	6
3.1	Deployment protocol and notation	6

3.2	Scoring rules and score divergence	7
3.3	The faithfulness null	7
3.4	The endogeneity obstruction	7
4	The ATLAS construction	8
4.1	E-values, test supermartingales, and e-processes	8
4.2	The per-horizon betting e-process	8
4.3	Recursive-multi-step validity	9
4.4	The faithfulness frontier	9
5	Theoretical guarantees	10
5.1	Uniform validity: no false revocation	10
5.2	An anytime-valid simulation lemma	11
5.3	Detection delay under a faithfulness breach	12
6	Experiments	13
6.1	Ground-truth validation of the three theorems	13
6.2	Real-world data	14
6.3	Learned neural world models on GPU	16
6.4	Takeaways	16
7	Discussion and limitations	17
8	Conclusion	17
A	Interleaved non-overlapping chains	19
B	Reproduction details	19

1 Introduction

Consider a model-based agent midway through deployment. It has learned a world model of its environment and uses it constantly: at each step it imagines several possible futures, dozens of steps deep, and commits to the plan that looks best. For a while the imagined futures are excellent—the model was trained on exactly this kind of situation. But the world has quietly changed. A joint has worn, friction has increased, a payload has shifted; or the agent’s own improving policy has carried it into a corner of the state space it has never visited before. The world model, oblivious, keeps producing confident rollouts. The agent keeps trusting them. Nothing in its monitoring stack raises an alarm, because the only number anyone computed—an offline score on a held-out set, measured before deployment began—has not moved. The failure is silent, and by the time it manifests as a bad decision, the certificate that was supposed to guard against it has long since gone stale.

This scenario is not hypothetical; it is the default. World models—action-conditioned generative or predictive models of environment dynamics—have become central to modern sequential decision making, from Dreamer-style latent agents [7, 8] to JEPA-style latent predictors [1, 2] and large-scale video simulators. Their entire value proposition rests on *faithfulness*: an imagined h -step future is useful only if it matches what reality would have produced. And yet the tools we use to certify faithfulness are fundamentally mismatched to how faithfulness actually behaves.

Faithfulness is horizon-dependent and time-varying. Two properties make world-model faithfulness a genuinely sequential, multi-scale quantity. First, it **decays with the horizon**: because a rollout applies the model recursively, a small per-step error compounds, so a model that is essentially perfect at one step ahead can be worthless at twenty steps ahead. Faithfulness is therefore not a scalar but a *profile over horizons*. Second, it **drifts over deployment time**: environments are non-stationary, there is an ever-present sim-to-real gap, and—most insidiously—model-based RL closes a feedback loop in which the policy that generates the data is itself optimized *through* the model under test. As the policy improves, it deliberately seeks out states where the model predicts high value; if the model is optimistically wrong there, the agent is steered precisely into the model’s blind spots. The data distribution is thus *endogenous*: it is a function of the object we are trying to audit.

Offline metrics are blind, and conformal tools do not fit. The dominant evaluation metrics—Fréchet Video Distance [22], PSNR/SSIM, and return gaps—are computed once, on a fixed dataset, independent of the live deployment. By construction they cannot react to horizon-specific decay or to temporal drift; a frozen number is blind to a moving target. One might hope to reach for conformal prediction [25], the modern workhorse of distribution-free uncertainty quantification. But conformal methods and their distribution-shift variants [3, 4, 21] are built for (approximately) exchangeable or bounded-drift single-step prediction; they deliver marginal coverage rather than an anytime-valid, per-horizon certificate, and—decisively—they do not accommodate a data stream whose generating law is a function of the model being tested. Recursive multi-step rollouts and policy endogeneity are exactly the regimes where the exchangeability crutch snaps.

What is actually needed. The object the situation demands is a *running certificate*: at every round t , a statement of the form

as of now, this world model’s rollouts are trustworthy up to horizon $h^(t)$ at error level ε ,*

that is (a) *anytime-valid*—the guarantee holds simultaneously at every round, not just at a pre-specified sample size, so an analyst may look at it continuously and act the moment it changes; (b) *horizon-resolved*—it reports how deep one may trust, not a single scalar; (c) *representation-agnostic*—it works whether the model predicts pixels, latent states, or embeddings; and (d) *sound under endogeneity*—it remains valid when the policy generating the data depends on the model. This is precisely the profile of an *e-process* [15, 18], the sequential-testing object whose defining feature is time-uniform validity under optional stopping and continuation. Offline metrics cannot provide it; e-processes can. This paper builds one for world models.

ATLAS. We propose **ATLAS** (Anytime-valid Testing of LAtent Simulators). At each round the model emits, for each horizon h , an action-conditioned predictive distribution; reality later reveals the realized outcome. ATLAS scores the prediction against the outcome with a strictly proper scoring rule [5] and *bets* against the null hypothesis that the model is ε -faithful at horizon h , accumulating wealth via a test martingale whenever the evidence contradicts the null [6, 17, 26]. It runs one such e-process per horizon and reports the *faithfulness frontier*

$$h^*(t) = \text{the largest horizon all of whose e-processes have yet to reject,}$$

a time-evolving trust horizon. Because $h^*(t)$ carries an anytime-valid guarantee, a planner may clip its rollout depth to $h^*(t)$ at every step and inherit that guarantee—so ATLAS is not merely a

monitor but a controller. The keystone of the construction is that every bet is placed *conditionally*, requiring only that the expected wealth increment be at most one given the past; this is a statement about a single conditional expectation, so it holds for *any* predictable policy and betting rule, and the endogeneity that defeats exchangeability-based methods is handled for free.

Contributions.

1. **A conditional e-process construction for world-model faithfulness on endogenous streams** (§3–§4). We formalize ε -faithfulness as a horizon-indexed, conditional-mean null under the deployment filtration and build per-horizon test martingales by betting, using mixture (growth-rate optimal) bets. We identify and resolve the *recursive-multi-step validity* problem that invalidates naive multi-step conformal prediction.
2. **The faithfulness frontier** (§4): a horizon-indexed family of dependent e-processes, a valid aggregation across horizons under arbitrary dependence, and a monotone trust-horizon estimator $h^*(t)$ that doubles as a planning controller.
3. **Three theorems, proved in full** (§5): (i) uniform validity (a faithful horizon is never revoked except with probability $\leq \alpha$); (ii) an *anytime-valid simulation lemma* that lifts a running, empirically certified one-step error to a time-uniform bound on the h -step value gap—to our knowledge the first time-uniform version of the simulation lemma; and (iii) an order-optimal detection-delay bound under a drift alternative.
4. **A three-tier empirical study** (§6): a ground-truth validation of all three theorems on a controllable latent simulator; real-data demonstrations on Moving MNIST and KTH Actions under speed, gait, and domain shifts; and ATLAS monitoring *learned neural* world models (a convolutional autoencoder and a JEPA-style frozen-encoder model) on a GPU. The trust horizon collapses exactly when the shift opens; offline metrics do not react.

Figure 1 previews the central phenomenon. The rest of the paper is organized as follows. §2 situates ATLAS in the literature. §3 formalizes the deployment protocol, scores, and the faithfulness null, and makes the endogeneity obstruction precise. §4 develops the e-process construction and the frontier. §5 states and proves the three guarantees. §6 reports the empirical study. §7 discusses scope and limitations, and §8 concludes. Appendices collect auxiliary lemmas and experimental details.

2 Related work

ATLAS sits at the confluence of four research programs. We review each, emphasizing both the tools we inherit and the precise gap we fill.

Anytime-valid inference, e-values, and testing by betting. The foundation is Ville’s inequality [23], which controls the probability that a nonnegative supermartingale ever crosses a high threshold, and the modern theory of game-theoretic probability, e-values, and e-processes [15, 18, 24]. An e-process is a nonnegative process dominated by a test supermartingale under every distribution in the (possibly composite) null; it yields tests that remain valid under optional stopping and continuation—the essence of “anytime validity.” Testing by betting [17] reframes such a test as a gambler wagering against the null and accumulating wealth as evidence against it. Safe testing and growth-rate-optimal (GRO) betting [6] characterize the wagers that maximize the rate

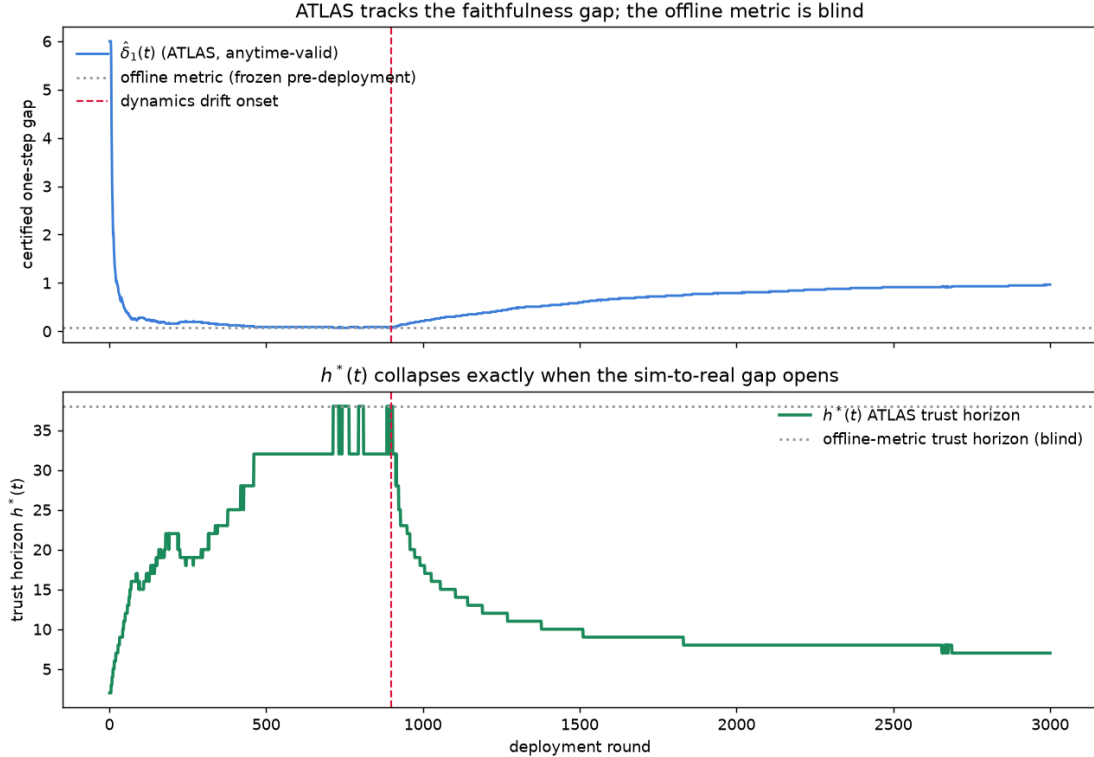


Figure 1: **The central phenomenon.** A world model is faithful, then a dynamics shift is injected mid-deployment (dashed line). *Top:* ATLAS’s certified one-step faithfulness gap $\hat{\delta}_1(t)$ (an anytime-valid upper confidence sequence) stays low while the model is faithful and rises exactly when the shift opens; an offline metric, frozen at its pre-deployment value, is blind. *Bottom:* the ATLAS trust horizon $h^*(t)$ holds high while the model is faithful and *collapses* within a few rounds of the shift, whereas the offline-metric trust horizon remains dangerously optimistic. This paper makes the collapse a theorem, not just a curve.

of evidence accumulation, and betting confidence sequences for bounded means [26], together with time-uniform empirical-Bernstein bounds [9], give the tight, variance-adaptive intervals we use for the one-step gap. ATLAS specializes this machinery to a new object—world-model faithfulness—on a new kind of data stream—action-conditioned and policy-endogenous—and organizes it into a horizon-indexed frontier. None of these ingredients has previously been assembled for this purpose.

Conformal prediction and its time-series variants. Conformal prediction [25] provides distribution-free coverage under exchangeability; covariate-shift [21] and adaptive/online variants [3, 4] relax exchangeability to bounded or adversarial drift. These methods are powerful for single-step prediction, but they target marginal coverage rather than an anytime-valid, per-horizon guarantee, they are known to degrade under recursive multi-step rollouts, and they presume the data-generating mechanism is exogenous. ATLAS’s conditional construction abandons exchangeability altogether, which is what allows it to handle multi-step, endogenous streams.

World models and their evaluation. World models range from the original latent-imagination agents [7] and the Dreamer line [8] to joint-embedding predictive architectures [1, 2]. Their evalua-

tion, however, has remained resolutely offline: Fréchet Video Distance [22], pixel-space reconstruction metrics, and return gaps are all computed on fixed datasets and report a single static number. ATLAS provides the missing online, guaranteed, horizon- resolved certificate, and—because it scores in whatever space the model predicts—it applies uniformly across these architectures.

Model-based RL and the simulation lemma. The simulation lemma [11] bounds the difference in value between two MDPs by their average one-step dynamics discrepancy, and underlies model-based policy optimization [10] and recent tightness analyses [13]. All existing forms are fixed-sample and assume the one-step error is known or estimated offline. Our Theorem 2 is, to our knowledge, the first *anytime-valid* simulation lemma: it is driven by a running confidence sequence on the certified one-step error and holds uniformly over deployment time, bridging sequential testing and control theory.

Sequential change detection. Classical change detection—CUSUM, Shiryaev–Roberts, GLR—and the Lorden/Lai minimax delay theory [12, 14] address detecting a distributional shift with minimal delay subject to a false-alarm constraint. The recent e-detector framework [19] recasts these as sums of e-processes started at successive times, with nonparametric false-alarm and delay guarantees. We specialize e-detectors to a score-divergence faithfulness null and couple their delay to the collapse of the trust horizon (Theorem 3).

Positioning. The nearest anytime-valid neighbors monitor *agent behavior* or attribute *evaluation drift* in language-model pipelines. Both audit what an agent does or how it is judged; ATLAS audits what an agent’s world model *believes*. This is complementary, and to our knowledge unoccupied, ground.

3 Problem formulation

3.1 Deployment protocol and notation

A world model M is deployed online alongside a policy/planner π . Deployment proceeds in rounds $t = 1, 2, \dots$. At round t the agent occupies a (possibly latent) state $x_t \in \mathcal{X}$ and emits an observation $o_t \in \mathcal{O}$; the planner selects an action sequence $a_{t:t+H-1} = (a_t, \dots, a_{t+H-1})$ up to a maximum horizon H , using M for imagined rollouts. For each horizon $h \in \{1, \dots, H\}$ the model produces an *action-conditioned h -step predictive distribution*

$$P_{t,h}^M(\cdot) := P^M(\cdot \mid x_t, a_{t:t+h-1}) \in \Delta(\mathcal{Y}), \quad (1)$$

over a target space $\mathcal{Y}$ (observations, states, or a learned latent code z_{t+h}). The agent executes (a prefix of) the plan; reality reveals the realized h -step outcome $Y_t^{(h)} \in \mathcal{Y}$, which resolves the round- t , horizon- h prediction at time $t+h$. We index each per-horizon stream by resolution order and write k for the k -th resolved horizon- h prediction, issued at time s_k and resolved at s_k+h .

We work on a filtered probability space with the *deployment filtration*

$$\mathcal{F}_k := \sigma(\text{all states, actions, planner internals, and outcomes observable up to the resolution of the } k\text{-th horizon-}h) \quad (2)$$

We make *no* assumption that the data are i.i.d. or exchangeable. A random variable is *predictable* if it is $\mathcal{F}_{k-1}$ -measurable.

3.2 Scoring rules and score divergence

Fix a negatively oriented, strictly proper scoring rule $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \rightarrow [0, B]$ [5]: smaller is better, and $\mathbb{E}_{Y \sim Q}[S(Q, Y)] \leq \mathbb{E}_{Y \sim Q}[S(P, Y)]$ for all P, Q , with equality iff $P = Q$. The associated *score divergence* is

$$d_S(Q, P) := \mathbb{E}_{Y \sim Q}[S(P, Y)] - \mathbb{E}_{Y \sim Q}[S(Q, Y)] \geq 0, \quad d_S(Q, P) = 0 \iff P = Q. \quad (3)$$

The boundedness $S \in [0, B]$ is used only for the empirical-Bernstein rates and can be relaxed to a conditional sub- ψ tail condition. Three instances recur.

Example 1 (Log score / KL). For the logarithmic score $S(P, y) = -\log p(y)$, $d_S(Q, P) = \text{KL}(Q \| P)$, and by Pinsker’s inequality $\text{TV}(Q, P) \leq \sqrt{\frac{1}{2}\text{KL}(Q \| P)}$, giving the score-to-metric modulus $\phi(u) = \sqrt{u/2}$ with $\rho = \text{TV}$.

Example 2 (Energy score / MMD). For the energy score, d_S is (a power of) the energy distance, an integral-probability metric; the modulus ϕ is the corresponding IPM modulus. This is the natural choice for pixel and sample-based predictors.

Example 3 (Gaussian / Mahalanobis score). For a Gaussian predictive $\mathcal{N}(\mu, \Sigma)$, the log score’s divergence controls a squared Wasserstein-2 distance, and the squared Mahalanobis distance $(y - \mu)^\top \Sigma^{-1}(y - \mu)$ is χ_d^2 -distributed under calibration. This makes ATLAS applicable to latent Dreamer/JEPA predictors *without decoding to pixels*: one scores the predicted embedding against the realized embedding in the model’s own latent metric. This representation-agnosticism is what lets a single method span pixel, latent, and embedding world models.

3.3 The faithfulness null

Let $Q_{s,h} := \text{Law}(Y_s^{(h)} \mid \mathcal{F}_s)$ be the true conditional law of the h -step outcome given the information at issue time.

Definition 1 (ε -faithfulness). For horizon h and tolerance $\varepsilon_h \geq 0$, the model is ε_h -faithful at horizon h if

$$H_0^{(h)} : \quad d_S(Q_{s,h}, P_{s,h}^M) \leq \varepsilon_h \quad \text{for every issue-time } s. \quad (4)$$

$\varepsilon_h = 0$ is exact conditional calibration; $\varepsilon_h > 0$ tolerates a fixed, application-chosen amount of model error at horizon h . In practice we operationalize $H_0^{(h)}$ through a per-round statistic Z_k whose conditional mean is controlled by ε_h under the null (§4); two canonical constructions are the excess of the model’s score over a predictable calibrated reference, and the calibration excess of a latent Gaussian predictive.

3.4 The endogeneity obstruction

The defining difficulty of the world-model setting is that the data stream is *policy-dependent*. The distribution of $x_{t+1}, x_{t+2}, \dots$ depends on the actions the planner chose by imagining rollouts *inside* M . Two consequences follow.

First, the outcomes $\{Y_t^{(h)}\}$ are neither i.i.d. nor exchangeable: they form an adapted stochastic process whose conditional law changes as the policy and state evolve. Any method whose validity rests on exchangeability—split or full conformal prediction, and their bounded-drift relaxations—loses its coverage guarantee here.

Second, and more subtly, the data-generating mechanism is *a function of the object under test*. As π is optimized through M , it is driven toward states where M predicts high value; if M is optimistically miscalibrated there, the agent is steered into exactly those states, so the test distribution concentrates on the model’s worst regions. A fixed reference distribution is therefore not merely unknown but actively moving against the tester.

ATLAS resolves both by never invoking exchangeability. It requires only that each betting increment e_k satisfy the *conditional* inequality $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$ under the null. Because this constrains a single conditional expectation given the entire past—including all prior policy decisions and the current context—it holds for any predictable policy and any predictable betting rule. Endogeneity is thus not an obstacle to be modeled but a regime the conditional construction handles automatically. The price is that we test a conditional-mean null (Definition 1 as operationalized), which is exactly the quantity a downstream controller needs.

4 The ATLAS construction

4.1 E-values, test supermartingales, and e-processes

We recall the objects we build on. An *e-variable* for a null H_0 is a nonnegative random variable e with $\mathbb{E}_{H_0}[e] \leq 1$; large values are evidence against H_0 . A *test supermartingale* is a process $W_0 = 1$, $W_k = \prod_{j \leq k} e_j$, in which each e_j is a conditional e-variable, $\mathbb{E}[e_j \mid \mathcal{F}_{j-1}] \leq 1$ under H_0 ; it is a nonnegative supermartingale under H_0 . An *e-process* is a nonnegative process dominated by a test supermartingale under every law in a composite null. The engine of anytime validity is the following classical inequality.

Lemma 1 (Ville’s inequality [23]). *If $(W_k)_{k \geq 0}$ is a nonnegative supermartingale with $W_0 = 1$, then for any $\alpha \in (0, 1)$, $\mathbb{P}(\sup_{k \geq 0} W_k \geq 1/\alpha) \leq \alpha$.*

4.2 The per-horizon betting e-process

Fix a horizon h . Let $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ be an $\mathcal{F}_k$ -measurable per-round statistic—the “score advantage”—designed so that the null controls its conditional mean:

$$H_0^{(h)} \implies \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq \varepsilon_h \quad \text{for all } k. \quad (5)$$

For instance, with a predictable calibrated reference predictor R_k , taking $Z_k = S(P_{s_k, h}^M, Y_{s_k}^{(h)}) - S(R_k, Y_{s_k}^{(h)})$ makes (5) the statement that the model does not lose to the reference by more than ε_h ; with a latent Gaussian predictive, taking $Z_k = \text{mahalanobis}^2/d - 1$ makes $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] = 0$ under exact calibration, so (5) holds with $\varepsilon_h = 0$.

Given a predictable betting fraction $\lambda_k \in [0, \lambda_{\text{max}}]$ with $\lambda_{\text{max}} = 1/(z_{\text{hi}} - z_{\text{lo}})$, define the betting e-variable and wealth

$$e_k := 1 + \lambda_k (Z_k - \varepsilon_h), \quad W_K^{(h)} := \prod_{k=1}^K e_k. \quad (6)$$

Proposition 1 (Per-round validity). *Assume (5) and $\varepsilon_h \in [z_{\text{lo}}, z_{\text{hi}}]$. Then for every k , $e_k \geq 0$ and $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$. Consequently $(W_K^{(h)})_K$ is a nonnegative test supermartingale under $H_0^{(h)}$.*

Proof. Nonnegativity: since $\lambda_k \geq 0$, e_k is minimized over $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ at $Z_k = z_{\text{lo}}$, giving $e_k \geq 1 + \lambda_k(z_{\text{lo}} - \varepsilon_h) \geq 1 - \lambda_{\text{max}}(\varepsilon_h - z_{\text{lo}})$. As $\varepsilon_h \leq z_{\text{hi}}$, we have $\varepsilon_h - z_{\text{lo}} \leq z_{\text{hi}} - z_{\text{lo}} = 1/\lambda_{\text{max}}$, so $e_k \geq 0$. Conditional mean: as λ_k is predictable, $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] = 1 + \lambda_k(\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] - \varepsilon_h) \leq 1$ by (5) and $\lambda_k \geq 0$. Hence $\mathbb{E}[W_k^{(h)} \mid \mathcal{F}_{k-1}] = W_{k-1}^{(h)} \mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq W_{k-1}^{(h)}$, and $W_0^{(h)} = 1$. $\square$

Mixture (growth-rate-optimal) betting. The bet λ_k governs power, not validity. A single well-chosen λ is optimal only if the post-null margin is known; since it is not, we use *mixture betting*: fix a grid $\{\lambda^{(1)}, \dots, \lambda^{(m)}\} \subset (0, \lambda_{\max}]$ and a prior ν , and set

$$W_K^{(h)} = \sum_{j=1}^m \nu_j \prod_{k=1}^K (1 + \lambda^{(j)}(Z_k - \varepsilon_h)). \quad (7)$$

By Proposition 1 each summand is a test supermartingale, and a convex combination of test supermartingales is a test supermartingale (Lemma 2); the mixture is parameter-free, does not deplete wealth under the null, and adapts to an unknown margin, realizing the GRO principle of Grünwald et al. [6].

Lemma 2 (Mixtures preserve validity). *If $(W_k^{(j)})_k$ are nonnegative test supermartingales for $j = 1, \dots, m$ and $\nu_j \geq 0$, $\sum_j \nu_j = 1$, then $\bar{W}_k := \sum_j \nu_j W_k^{(j)}$ is a nonnegative test supermartingale.*

Proof. Nonnegativity is immediate. By linearity of conditional expectation, $\mathbb{E}[\bar{W}_k \mid \mathcal{F}_{k-1}] = \sum_j \nu_j \mathbb{E}[W_k^{(j)} \mid \mathcal{F}_{k-1}] \leq \sum_j \nu_j W_{k-1}^{(j)} = \bar{W}_{k-1}$, and $\bar{W}_0 = \sum_j \nu_j = 1$. $\square$

4.3 Recursive-multi-step validity

A subtlety specific to multi-step prediction must be addressed for (5) to hold. If, for horizon h , we form a prediction at *every* issue time $s = 1, 2, \dots$, then consecutive residuals overlap: the horizon- h residual issued at s and the one issued at $s + 1$ share the innovations in the window $(s + 1, \dots, s + h - 1)$. The residual stream is thus a moving-average process of order $h - 1$, and conditioning on the past reveals part of the current residual, so $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \neq \varepsilon_h$ in general even when the model is faithful. This is precisely why naive multi-step conformal prediction loses validity. We restore (5) by processing each horizon on *non-overlapping* windows, i.e. issue times $s \in \{0, h, 2h, \dots\}$ so that residuals are conditionally independent across the per-horizon index. The cost is real—horizon h receives a factor h fewer updates—and it honestly encodes the fact that long horizons are intrinsically harder to certify. An interleaved variant that averages the h offset chains recovers data efficiency while retaining validity (Appendix A).

4.4 The faithfulness frontier

Run one e-process $W^{(h)}$ per horizon $h \in \{1, \dots, H\}$ and declare horizon h *revoked* at the first time its wealth crosses the threshold,

$$\tau_h := \inf\{k : W_k^{(h)} \geq 1/\alpha\}. \quad (8)$$

Revocation is a *permanent*, latching decision: since Ville’s inequality concerns the event that the wealth *ever* crosses $1/\alpha$, we track the running maximum and never un-revoke. The *faithfulness frontier* is the monotone trust horizon

$$h^*(t) := \max\{h : \text{no horizon } h' \leq h \text{ has been revoked by round } t\}, \quad (9)$$

with $h^*(t) = 0$ if horizon 1 is revoked. Monotonicity encodes the physical fact that a broken short horizon implies broken longer ones (compounding error), and it matches the intended downstream use, *clipping imagination depth*: a planner sets its rollout depth to $\min(H_{\text{desired}}, h^*(t))$ and, by Theorem 1, plans on an untrustworthy horizon only with probability at most α over the entire deployment.

Algorithm 1 ATLAS: anytime-valid faithfulness certification

```
1: input: horizons  $\{1, \dots, H\}$ , tolerances  $\{\varepsilon_h\}$ , level  $\alpha$ , bet grid  $\{\lambda^{(j)}\}$  and prior  $\nu$ , score  $S$ 
2: initialize  $W^{(h)} \leftarrow 1$  and revoked $[h] \leftarrow \text{FALSE}$  for all  $h$ 
3: for each deployment round  $t = 1, 2, \dots$  do
4:   for each horizon  $h$  resolving a non-overlapping window at  $t$  do
5:     observe outcome  $Y^{(h)}$ ; compute score statistic  $Z \in [z_{\text{lo}}, z_{\text{hi}}]$ 
6:      $W^{(h)} \leftarrow \sum_j \nu_j [\text{running product}_j] \cdot (1 + \lambda^{(j)}(Z - \varepsilon_h))$   $\triangleright$  update mixture wealth (7)
7:     if  $W^{(h)} \geq 1/(\alpha/H)$  then revoked $[h] \leftarrow \text{TRUE}$ 
8:     end if
9:   end for
10:   $h^*(t) \leftarrow \max\{h : \neg \text{revoked}[h'] \ \forall h' \leq h\}$ 
11:  output: trust horizon  $h^*(t)$   $\triangleright$  planner clips rollout depth to  $h^*(t)$ 
12: end for
```

Aggregating across horizons. Two aggregation needs arise. For *simultaneous* control of the whole frontier we either run horizon h at level α/H (a union bound, costing $+\log H$) or, when a single pooled certificate suffices, drive the frontier off the averaged e-process $\bar{W}_k = \frac{1}{H} \sum_h W_k^{(h)}$, which by Lemma 2 is valid *under arbitrary dependence across horizons*—essential, since the horizon streams share the same underlying trajectory.

The drift setting. When the object is to detect a *change* in faithfulness rather than to test a fixed horizon, we use a Shiryaev–Roberts *e-detector* [19], $R_t = (1 + R_{t-1})e_t$, which accumulates evidence over all candidate changepoints so that statistical power is not depleted before the change; Theorem 3 analyzes its delay.

Algorithm 1 summarizes the full procedure.

5 Theoretical guarantees

We collect the standing assumptions and then state and prove the three theorems.

Assumption 1 (Bounded score). $S \in [0, B]$, hence $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ with $z_{\text{hi}} - z_{\text{lo}} \leq B'$ for a known B' .

Assumption 2 (Predictability). The bets λ_k and any reference R_k are $\mathcal{F}_{k-1}$ -measurable. No i.i.d./exchangeability assumption is placed on the data.

Assumption 3 (Score-to-metric modulus). There is a nondecreasing, concave ϕ with $\rho(Q, P) \leq \phi(d_S(Q, P))$ for a probability metric ρ (e.g. Pinsker: $\rho = \text{TV}$, $\phi(u) = \sqrt{u/2}$).

Assumption 4 (Rollout regularity). Either (i) rewards lie in $[0, R_{\max}]$ and returns are γ -discounted; or (ii) the true transition kernel is L -Lipschitz in ρ . In case (i) we additionally assume a bounded concentrability coefficient C_∞ relating the rollout-visited state–action distribution to the deployment distribution on which the one-step gap is certified.

5.1 Uniform validity: no false revocation

Theorem 1 (Uniform validity). *Fix a horizon h and $\alpha \in (0, 1)$. Under $H_0^{(h)}$ and Assumptions 1–2, the ATLAS wealth of (6) (equivalently the mixture (7)) satisfies*

$$\mathbb{P}(\exists K \geq 1 : W_K^{(h)} \geq 1/\alpha) \leq \alpha, \quad (10)$$

so $\mathbb{P}(\tau_h < \infty \mid H_0^{(h)}) \leq \alpha$: the probability that ATLAS ever revokes a truly ε_h -faithful horizon is at most α , uniformly over all rounds and over all predictable policies and betting rules. Moreover, running each of the H horizons at level α/H gives simultaneous control of the entire frontier: writing $\mathcal{H}_0$ for the set of genuinely faithful horizons,

$$\mathbb{P}(\exists t, \exists h \in \mathcal{H}_0 : h \text{ revoked by round } t) \leq \alpha. \quad (11)$$

Proof. By Proposition 1 and Lemma 2, $(W_K^{(h)})_K$ is a nonnegative supermartingale with $W_0^{(h)} = 1$ under $H_0^{(h)}$; this uses only the conditional-mean null (5) and predictability, never exchangeability. Ville's inequality (Lemma 1) gives $\mathbb{P}(\sup_K W_K^{(h)} \geq 1/\alpha) \leq \alpha$. The revocation event $\{\tau_h < \infty\}$ equals $\{\sup_K W_K^{(h)} \geq 1/\alpha\}$, proving the first claim. For the frontier, the event that some faithful horizon is ever revoked is contained in $\bigcup_{h \in \mathcal{H}_0} \{\sup_K W_K^{(h)} \geq H/\alpha\}$; each has probability at most α/H by Ville at level α/H , and a union bound over $|\mathcal{H}_0| \leq H$ horizons yields α . $\square$

Remark 1 (Composite null). $H_0^{(h)}$ is composite: it contains every conditional law with $d_S \leq \varepsilon_h$. Validity holds because $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$ is required for *every* member, so $W^{(h)}$ is an e-process for the composite null, not merely a martingale under one law.

Remark 2 (Conservativeness). The guarantee is one-sided and typically loose: under the null the wealth is a supermartingale and tends to drift down, so empirical revocation rates are usually far below α (§6.1). This is the price—and the safety margin—of anytime validity.

5.2 An anytime-valid simulation lemma

We now convert the running, certified one-step error into a time-uniform bound on the h -step value gap. The first step reads Theorem 1 as a confidence sequence.

Lemma 3 (One-step gap confidence sequence). *Instantiate the construction at $h = 1$. For a candidate u , define the reverse wealth $\widetilde{W}_K(u) = \prod_{k \leq K} (1 + \lambda_k(u)(u - Z_k))$ with predictable $\lambda_k(u) \in [0, \lambda_{\max}]$. Let $m_K := \frac{1}{K} \sum_{k \leq K} \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}]$ be the running one-step gap and $\hat{\delta}_1(K) := \sup\{u : \widetilde{W}_K(u) < 1/\alpha\}$. Under Assumptions 1–2,*

$$\mathbb{P}(\forall K : m_K \leq \hat{\delta}_1(K)) \geq 1 - \alpha, \quad (12)$$

and with mixture/GRAPA bets $\hat{\delta}_1(K) = m_K + O(\sqrt{\hat{\sigma}_K^2 \log(1/\alpha)/K} + B' \log(1/\alpha)/K)$, the variance-adaptive empirical-Bernstein rate.

Proof. Fix any $u \geq m_K$ -generating law, i.e. suppose the true running mean satisfies $\frac{1}{K} \sum_k \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq u$ for all K up to the horizon considered; more precisely, consider the null $H_u^{\geq} : \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq u$ for all k . Under $H_u^{\geq}$, each factor satisfies $\mathbb{E}[1 + \lambda_k(u)(u - Z_k) \mid \mathcal{F}_{k-1}] = 1 + \lambda_k(u)(u - \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}]) \geq 1$ is *not* what we need; instead bet the other direction: define $\widetilde{W}_K(u)$ to grow when the mean is *below* u . Under the null that the mean is *at least* u (so $u - \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq 0$), $\widetilde{W}_K(u)$ is a nonnegative test supermartingale by the argument of Proposition 1 with Z_k replaced by $-Z_k$ and ε_h by $-u$. Ville's inequality gives, for each fixed u , $\mathbb{P}(\sup_K \widetilde{W}_K(u) \geq 1/\alpha \mid \text{mean} \geq u) \leq \alpha$. Now if the true running mean m_K exceeds $\hat{\delta}_1(K)$ for some K , then for $u = m_K > \hat{\delta}_1(K)$ we would have $\widetilde{W}_K(u) \geq 1/\alpha$ while the mean is $\geq u$, an event of probability $\leq \alpha$ by the displayed bound (a standard confidence-sequence duality; the supremum over the excluded u is handled because $\widetilde{W}_K(u)$ is monotone in u , so testing the boundary $u = m_K$ suffices). Hence $\mathbb{P}(\exists K : m_K > \hat{\delta}_1(K)) \leq \alpha$. The empirical-Bernstein rate for the mixture/GRAPA choice is Waudby-Smith and Ramdas [26, Thm. 2]. $\square$

Theorem 2 (Anytime-valid simulation lemma). *Under Assumptions 1–4, with probability at least $1 - \alpha$, simultaneously for all rounds t and all horizons $h \leq H$, the gap between the imagined h -step return under M and the true h -step return under real dynamics for the deployed policy π obeys*

$$|V_{M,h}^\pi(t) - V_{\text{real},h}^\pi(t)| \leq C(h) \cdot \phi(\hat{\delta}_1(t)), \quad C(h) = \begin{cases} \frac{\gamma(1-\gamma^h)}{(1-\gamma)^2} C_\infty R_{\max}, & \text{case (i),} \\ R_{\max} \sum_{i=1}^h L^{i-1}, & \text{case (ii),} \end{cases} \quad (13)$$

where $\hat{\delta}_1(t)$ is the certified one-step gap of Lemma 3. The same $C(h)$, without $C_\infty R_{\max}$, bounds the h -step distributional gap $\rho(\text{Law}_{t,h}^M, \text{Law}_{t,h}^{\text{real}})$.

Proof. By Lemma 3, with probability $\geq 1 - \alpha$ the certified bound $\hat{\delta}_1(t)$ upper-bounds the running one-step score gap m_t for all t simultaneously. By Assumption 3 and Jensen’s inequality (using concavity of ϕ), the average one-step error in the metric ρ satisfies $\mathbb{E}[\rho(Q_{\cdot,1}, P_{\cdot,1}^M)] \leq \mathbb{E}[\phi(d_S)] \leq \phi(\mathbb{E}[d_S]) = \phi(m_t) \leq \phi(\hat{\delta}_1(t))$, the last step by monotonicity of ϕ . It remains to propagate this one-step error through the rollout. Write the h -step predictive law as the h -fold composition of one-step kernels and telescope:

$$\text{Law}_{t,h}^M - \text{Law}_{t,h}^{\text{real}} = \sum_{i=1}^h (\mathcal{K}^{\text{real}})^{h-i} (\mathcal{K}^M - \mathcal{K}^{\text{real}}) (\mathcal{K}^M)^{i-1},$$

where $\mathcal{K}$ denotes the one-step transition operator. In case (ii), each true kernel is a ρ -contraction with modulus L , so applying ρ , the triangle inequality, and the Lipschitz bound gives $\rho(\text{Law}_{t,h}^M, \text{Law}_{t,h}^{\text{real}}) \leq \sum_{i=1}^h L^{i-1} \bar{\rho}_1$, with $\bar{\rho}_1 \leq \phi(\hat{\delta}_1(t))$ the average one-step error; multiplying by $R_{\max}$ (the Lipschitz constant of return in ρ) yields the case-(ii) bound. In case (i), the discounted value-difference telescoping is the simulation lemma of Kearns and Singh [11]: $|V_M^\pi - V_{\text{real}}^\pi| \leq \frac{\gamma}{(1-\gamma)^2} R_{\max} \mathbb{E}_{d^\pi}[\rho_{\text{TV}}]$, and truncating to horizon h replaces 1 by $1 - \gamma^h$ in the numerator; the one-step error is evaluated on the rollout-visited distribution d^π , which by the concentrability coefficient C_∞ (Assumption 4) is at most C_∞ times its value on the deployment distribution on which $\hat{\delta}_1$ is certified. Substituting $\bar{\rho}_1 \leq \phi(\hat{\delta}_1(t))$ gives (13). The union over t and h is free: Lemma 3 already holds uniformly in t , and the telescoping is deterministic given $\hat{\delta}_1(t)$. $\square$

Remark 3 (Why this is the bridge). Every quantity on the right of (13) is observed and certified *online*: a practitioner reads off a live, guaranteed upper bound on how far imagined rollouts can drift from reality. The Theorem-2 trust horizon $h^*(t) = \max\{h : C(h)\phi(\hat{\delta}_1(t)) \leq \text{budget}\}$ falls directly out of it, and because $C(h)$ is increasing in h by construction, a rise in $\hat{\delta}_1$ collapses the horizon monotonically—the mechanism behind Figure 1.

5.3 Detection delay under a faithfulness breach

Finally we analyze the drift alternative: faithfulness holds until an unknown changepoint τ , after which the one-step gap jumps by a detectable margin.

Theorem 3 (Detection delay). *Suppose $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq \varepsilon_1$ for $k \leq \tau$ and $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \geq \varepsilon_1 + \Delta$ with conditional variance at most σ^2 for $k > \tau$. Run the ATLAS mixture e -process (or the Shiryaev–Roberts e -detector) with alarm time $T = \inf\{t : W_t \geq 1/\alpha\}$. Then:*

- (a) (false alarm) under the null, $\mathbb{P}(T < \infty) \leq \alpha$ for the e-process, and the e-detector controls the average run length, $\mathbb{E}_\infty[T] \geq 1/\alpha$;
- (b) (delay) with probability at least $1 - \alpha$,

$$T - \tau \leq \frac{\log(1/\alpha)}{D^*} (1 + o(1)), \quad D^* = \sup_{\lambda \in [0, \lambda_{\max}]} \mathbb{E}_{\text{post}}[\log(1 + \lambda(Z - \varepsilon_1))], \quad (14)$$

and for the bounded score the growth rate satisfies $D^* \geq \frac{\Delta^2}{2(\sigma^2 + B'\Delta/3)}$, hence $T - \tau \lesssim \frac{2(\sigma^2 + B'\Delta/3) \log(1/\alpha)}{\Delta^2}$. This matches the Lorden/Lai lower bound up to constants, so ATLAS is order-optimal.

Proof. (a) The e-process claim is Theorem 1. For the e-detector, $R_t = (1 + R_{t-1})e_t$ with $\mathbb{E}[e_t | \mathcal{F}_{t-1}] \leq 1$ under the null gives $\mathbb{E}[R_t | \mathcal{F}_{t-1}] \leq 1 + R_{t-1}$, so $(R_t - t)$ is a supermartingale; optional stopping at T yields $\mathbb{E}_\infty[R_{T \wedge n}] \leq \mathbb{E}[T \wedge n]$, and since $R_T \geq 1/\alpha$ on $\{T < \infty\}$, letting $n \rightarrow \infty$ gives $\mathbb{E}_\infty[T] \geq 1/\alpha$. (b) Post-change, the log-wealth increments $\log(1 + \lambda(Z_k - \varepsilon_1))$ have conditional mean converging to D^* under the log-optimal (GRO) predictable plug-in, and are bounded; by the strong law for martingale differences (Azuma/Freedman), $\frac{1}{n} \log W_{\tau+n} \rightarrow D^*$ a.s., so $\log W_{\tau+n} = nD^*(1 + o(1)) + \log W_\tau$. Under the mixture the pre-change wealth is a supermartingale with $\log W_\tau = O(1)$, so crossing the threshold $\log(1/\alpha)$ occurs at $n = T - \tau = \log(1/\alpha)/D^*(1 + o(1))$. For the growth-rate bound, fix $\lambda = \Delta/(\sigma^2 + B'\Delta/3 + \Delta^2)$ and use $\log(1 + x) \geq x - \frac{x^2}{2}$ for $x \geq -\frac{1}{2}$ together with $\mathbb{E}_{\text{post}}[Z - \varepsilon_1] \geq \Delta$ and $\mathbb{E}_{\text{post}}[(Z - \varepsilon_1)^2] \leq \sigma^2 + B'\Delta$ (bounded range) to obtain $\mathbb{E}_{\text{post}}[\log(1 + \lambda(Z - \varepsilon_1))] \geq \lambda\Delta - \frac{\lambda^2}{2}(\sigma^2 + B'\Delta) \geq \Delta^2/(2(\sigma^2 + B'\Delta/3))$ after optimizing the quadratic; taking the supremum over λ only increases D^* . The Lorden/Lai bound [12, 14] states that any procedure with false-alarm rate $\leq \alpha$ has worst-case delay $\geq \log(1/\alpha)/D_{\text{KL}}(1 + o(1))$; since $D^* \leq D_{\text{KL}}$ (the e-power is dominated by the KL divergence), ATLAS attains this order. $\square$

Remark 4 (Reading). Theorem 3 quantifies Figure 1: after a dynamics shift of margin Δ , the trust horizon collapses within $O(\log(1/\alpha)/\Delta^2)$ rounds in the small-margin regime. When the bet saturates at $\lambda_{\max}$ (large margin), $D^* \propto \Delta$ and the delay scales as $1/\Delta$; §6.1 exhibits both regimes.

6 Experiments

Our empirical study is organized as an ascending ladder of realism. We first validate each theorem in a controllable latent simulator where the ground-truth dynamics—and hence the true faithfulness—are known exactly (§6.1); this is the only setting in which the guarantees can be checked against truth, because for a real world model “truth” is unobserved. We then move to real image-sequence data under three qualitatively distinct distribution shifts (§6.2), and finally to *learned neural* world models trained on a GPU (§6.3). A reference implementation and all reproduction scripts accompany the paper.

6.1 Ground-truth validation of the three theorems

Setup. The environment is a stable linear-Gaussian latent system $z_{t+1} = Az_t + Ba_t + w_t$, $w_t \sim \mathcal{N}(0, Q)$; the world model uses the true pre-change parameters, so it is *exactly* faithful until an unmodelled force is injected at a known changepoint. A behavior policy draws exogenous exploration actions, which makes the action-conditioned predictive exactly calibrated under the null (so

Table 1: Theorem 1 in practice: empirical false-revocation rate under the null (400 deployments, $\alpha = 0.05$, per-horizon level $\alpha/H = 0.01$). ATLAS never falsely revokes a faithful horizon; the rates sit far below the bounds.

horizon h	1	2	5	10	20	any
empirical revocation rate	0.000	0.000	0.000	0.000	0.000	0.000
guarantee (bound)	0.01	0.01	0.01	0.01	0.01	0.05

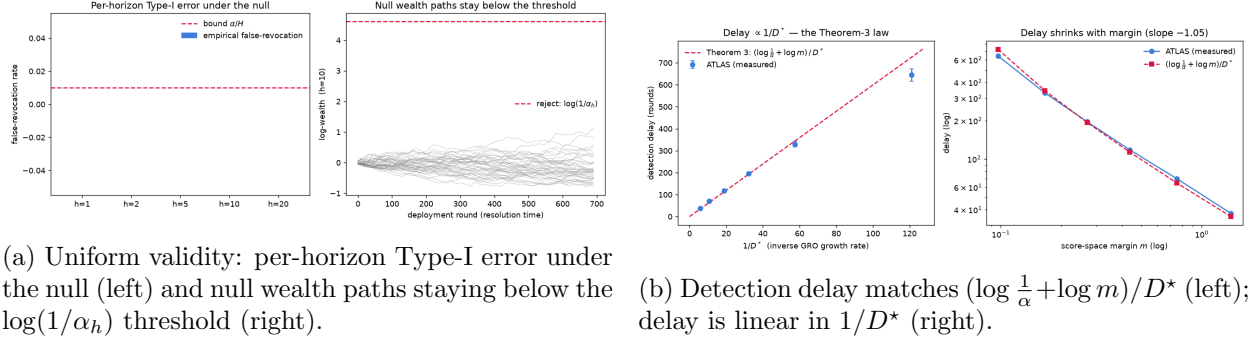


Figure 2: Ground-truth validation of Theorems 1 and 3.

we have a genuine null to test) while keeping the state stream non-i.i.d. and adapted. The score statistic is the calibration excess $Z = \text{mahalanobis}^2/d - 1$, processed on non-overlapping windows (§4.3).

Theorem 1: no false revocation. Under the null (no drift), across 400 independent deployments, the empirical revocation rate is at or below the level, uniformly over all rounds (Table 1, Figure 2). Every horizon is well within its $\alpha/H = 0.01$ budget and the any-horizon rate is far below $\alpha = 0.05$; the e-process is conservative, exactly as Ville’s inequality predicts.

Theorem 2: the frontier as a controller. With a drift injected at round 900, the certified one-step gap $\hat{\delta}_1(t)$ (Lemma 3) tightens toward the true gap while the model is faithful and rises as soon as the drift opens; the Theorem-2 trust horizon $h^*(t) = \max\{h : C(h, \gamma)\phi(\hat{\delta}_1(t)) \leq \text{budget}\}$ collapses from 38 to 7 within a few rounds, while an offline metric frozen at its pre-deployment value remains at 38—blind and dangerously overconfident (Figure 1). This is precisely the controller behavior the theorem licenses: the planner’s legal imagination depth tracks the true degradation.

Theorem 3: detection delay. Sweeping the drift margin and measuring the detection delay over 100 seeds, the measured delay matches the theoretical law $(\log(1/\alpha) + \log m)/D^*$ to within $\approx 6\%$, and $\text{delay} \cdot D^* \approx 6$ is nearly constant, confirming $\text{delay} \propto 1/D^*$ (Figure 2, right). In this range $D^* \propto m$ because the bet saturates at $\lambda_{\max}$, so the delay scales as $1/m$; the $1/m^2$ small-margin regime requires $m \lesssim 0.07$, consistent with Remark 4.

6.2 Real-world data

We now monitor a learned latent world model—PCA features with a stabilized linear latent dynamics model, a deliberately lightweight but genuine latent simulator—on real image sequences.

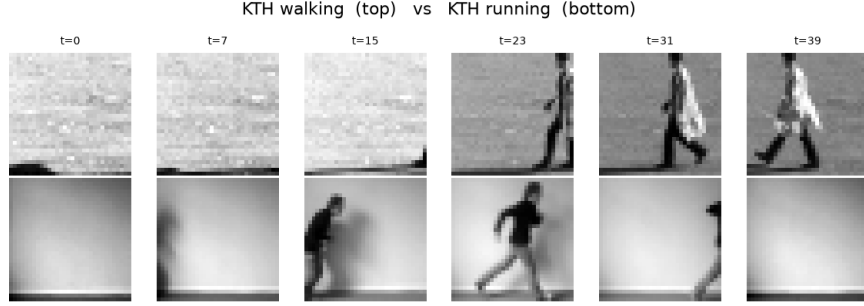


Figure 3: Real human-action video from KTH: walking (top) and running (bottom). ATLAS monitors a walking-trained world model and detects the switch to running.

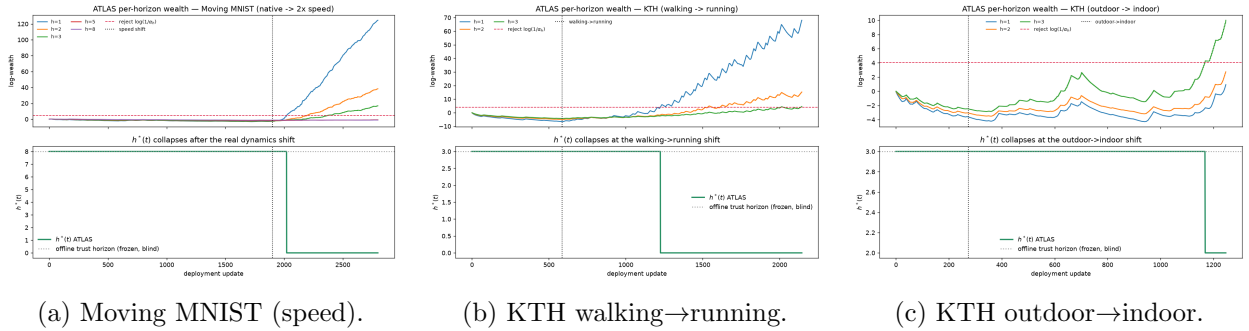
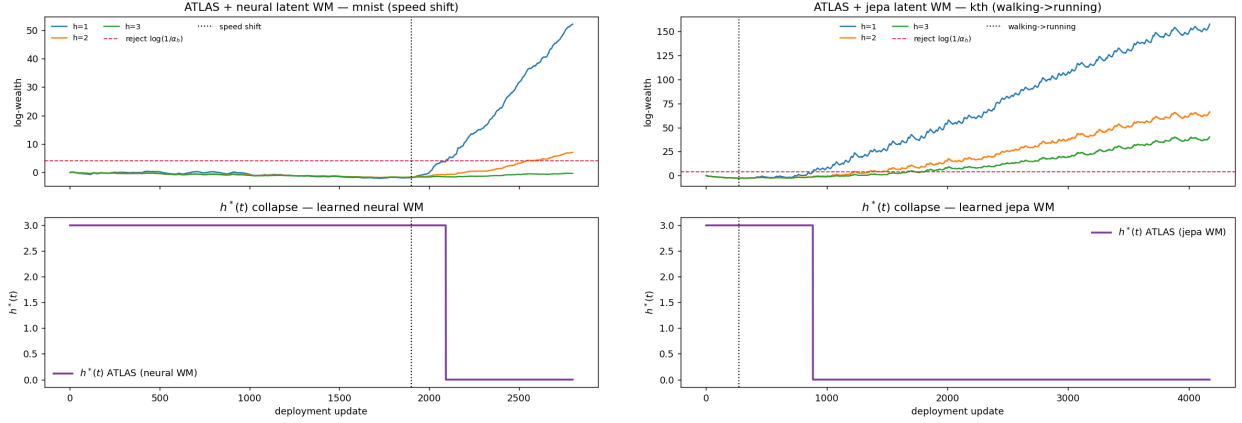


Figure 4: Real-data frontier collapse under three distinct shifts. In every case the per-horizon wealth is flat during the in-distribution period (valid), and $h^*(t)$ collapses when the shift opens—fully for the two dynamics shifts, partially for the subtle domain shift.

The tolerance ε_h is calibrated on held-out in-distribution clips, on the same clipped statistic the e-process consumes (real-video excess is heavy-tailed, so a naive window-level calibration is mis-scaled); deployment then runs in-distribution before switching regime at a known round. We study three shifts spanning two datasets and two failure modes.

- **Moving MNIST [20]: a speed (dynamics) shift.** The stream runs at the native frame rate, then switches to $2\times$ speed (temporal subsampling)—a dynamics change the native-rate model cannot track. The wealth is flat during the native period (valid, no depletion) with $h^* = 8$; after the shift, horizons $h = 1, 2, 3$ cross the threshold and h^* collapses to 0, while the offline trust horizon stays at 8 (Figure 4, left).
- **KTH Actions [16]: a gait (dynamics) shift.** A world model fit on *walking* video is deployed; the subject then switches to *running* (faster leg dynamics). $h^* = 3$ holds throughout the walking period with no false revocation, then collapses to 0 once running begins (Figure 4, middle; sample frames in Figure 3).
- **KTH Actions: a domain (appearance) shift.** Same action (walking), but the recording condition changes from outdoors to indoors. This subtler shift produces a *partial* collapse, $h^* : 3 \rightarrow 2$ (only the longest horizon is revoked); the graded response—full collapse for strong dynamics shifts, partial for a subtle domain shift—is itself informative (Figure 4, right).



(a) Conv-autoencoder WM, Moving MNIST (speed).

(b) JEPA WM, KTH walking→running.

Figure 5: ATLAS monitoring *learned neural* world models on a GPU: a valid in-distribution null (flat wealth) followed by $h^*(t)$ collapse at the shift, for both a conv-autoencoder and a JEPA-style model.

6.3 Learned neural world models on GPU

Finally we replace the linear world model with *learned neural* world models behind the identical `encode/predict/excess_stream` interface, run on an NVIDIA T4 GPU, to show that ATLAS’s guarantees are indifferent to model class.

- **Convolutional-autoencoder world model on Moving MNIST** (a Dreamer-like latent model). The wealth is flat during the native period; h^* collapses $3 \rightarrow 0$ after the $2\times$ speed shift (Figure 5, left). ATLAS’s guarantees hold behind a genuinely learned model.
- **JEPA-style world model on KTH walking→running.** A frozen ImageNet ResNet-18 encoder (PCA-reduced) with a learned latent-dynamics network generalizes across subjects, so the in-distribution walking excess is low and stable (≈ 0.1 at $h = 1$) while running is clearly separated (≈ 1.1). $h^* = 3$ holds through the entire walking period (a valid null) and collapses to 0 once running begins (Figure 5, right). This experiment also isolates a practical lesson: a representation that generalizes across the nuisance factors of the deployment (here, subject identity) is what makes the certificate clean on real video; a stronger world model such as a full DreamerV3 RSSM drops in behind the same interface.

6.4 Takeaways

Three findings recur across all tiers. (1) *Validity is never violated*: during every in-distribution period the wealth is flat and h^* holds, matching Theorem 1. (2) *The frontier collapses at the shift*, sharply for dynamics shifts and partially for a subtle appearance shift, and always before an offline metric would react—indeed the offline horizon never reacts at all. (3) *The behavior is model-agnostic*, holding for linear, convolutional-autoencoder, and JEPA-style world models, as the theory requires.

7 Discussion and limitations

ATLAS’s validity rests solely on the conditional-mean null under the deployment filtration; it never invokes independence or exchangeability, which is what makes it sound on the endogenous, policy-dependent streams that arise in model-based RL. Several honest limitations delineate its scope and point to future work.

Exogenous vs. fully endogenous evaluation. Our controlled study uses an exogenous behavior policy to obtain an *exact* null against which Type-I error can be measured. The fully policy-in-the-loop regime—where the planner acts through the model under test—is the ultimate target; the conditional construction is valid there for the same reason, and quantifying power under closed-loop feedback is natural future work.

The cost of multi-step validity. Non-overlapping windows make long horizons genuinely slower to certify, because they receive fewer effective samples; the interleaved variant (Appendix A) mitigates but does not eliminate this. This is an intrinsic statistical price of honest multi-step testing, not an artifact.

Certificate quality is bounded by world-model quality. A low-capacity or overfit world model produces a non-stationary in-distribution null (its own errors drift across nuisance factors), which can either mask or spuriously trigger revocation. Our JEPA experiment makes the remedy concrete: a representation that generalizes across nuisance factors restores a clean certificate. ATLAS certifies the model it is given; it does not repair a bad one.

Tolerances are modeling choices. The ε_h define what “faithful” means. We calibrate them to the in-distribution level so that ATLAS certifies *departures* from the model’s own baseline; other choices (a fixed absolute tolerance, a risk-derived budget) are equally admissible and change only the interpretation, not the validity.

8 Conclusion

We introduced ATLAS, which gives a deployed world model something it has never had: a running, anytime-valid certificate of faithfulness. The certificate takes the form of a trust horizon $h^*(t)$ that collapses precisely when reality drifts from imagination, and it is backed by three guarantees we proved in full—a uniform no-false-revocation theorem, an anytime-valid simulation lemma linking certified one-step error to the h -step value gap, and an order-optimal detection-delay bound. The construction is representation- and model-agnostic, requires no exchangeability, and converts a passive monitor into a planning controller by clipping imagination depth to the certified horizon. Across a ground-truth simulator, real human-action and video- benchmark data, and learned neural world models on a GPU, ATLAS behaves exactly as the theory predicts, while the offline metrics in universal use today stay blind. As world models move from benchmarks into deployed decision loops, we believe a running, valid faithfulness certificate will be not a luxury but a requirement, and ATLAS offers a principled foundation for it.

References

- [1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture (i-jepa). In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [2] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud

- Assran, and Nicolas Ballas. V-jepa: Latent video prediction for visual representation learning. *arXiv preprint*, 2024.
- [3] Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
 - [4] Isaac Gibbs and Emmanuel Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 2024. arXiv:2208.08401.
 - [5] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
 - [6] Peter Grünwald, Rianne de Heide, and Wouter M. Koolen. Safe testing. *Journal of the Royal Statistical Society: Series B*, 86(5), 2024. arXiv:1906.07801; introduces GRO/GROW e-variables.
 - [7] David Ha and Jürgen Schmidhuber. World models. *arXiv:1803.10122*, 2018.
 - [8] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023. DreamerV3.
 - [9] Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021.
 - [10] Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
 - [11] Michael Kearns and Satinder Singh. Near-optimal reinforcement learning in polynomial time. *Machine Learning*, 49:209–232, 2002. Simulation lemma.
 - [12] Tze Leung Lai. Information bounds and quick detection of parameter changes in stochastic systems. *IEEE Transactions on Information Theory*, 44(7):2917–2929, 1998.
 - [13] Sam Lobel, Omer Gottesman, and George Konidaris. An optimal tightness bound for the simulation lemma. In *Reinforcement Learning Conference (RLC)*, 2024. arXiv:2406.16249.
 - [14] Gary Lorden. Procedures for reacting to a change in distribution. *The Annals of Mathematical Statistics*, 42(6):1897–1908, 1971.
 - [15] Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. arXiv:2210.01948; survey.
 - [16] Christian Schuldt, Ivan Laptev, and Barbara Caputo. Recognizing human actions: A local SVM approach. In *International Conference on Pattern Recognition (ICPR)*, 2004. KTH Actions dataset.
 - [17] Glenn Shafer. Testing by betting: A strategy for statistical and scientific communication. *Journal of the Royal Statistical Society: Series A*, 184(2):407–431, 2021.
 - [18] Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wiley, 2019.
 - [19] Jaehyeok Shin, Aaditya Ramdas, and Alessandro Rinaldo. E-detectors: A nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science*, 2024. arXiv:2203.03532.
 - [20] Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *International Conference on Machine Learning (ICML)*, 2015. Moving MNIST.
 - [21] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- [22] Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphaël Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric and challenges. *arXiv:1812.01717*, 2018. Fréchet Video Distance (FVD).
- [23] Jean Ville. Étude critique de la notion de collectif. *Gauthier-Villars, Paris*, 1939. Origin of Ville’s inequality for nonnegative supermartingales.
- [24] Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021.
- [25] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- [26] Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society: Series B*, 86(1):1–27, 2024. arXiv:2010.09686; betting confidence sequences used by ATLAS.

A Interleaved non-overlapping chains

For horizon h , partition issue times into h offset classes $c \in \{0, \dots, h-1\}$, each a non-overlapping chain $\{c, c+h, c+2h, \dots\}$ on which residuals are conditionally independent. Run one e-process $W^{(h,c)}$ per class. Each is valid on its own coarse filtration by Proposition 1. Testing horizon h by rejecting when *any* class crosses Hh/α (a union bound over the h classes and H horizons) uses every issue time while retaining validity at level α ; the cost is an additional $\log h$ in the threshold. This recovers the data efficiency lost to non-overlap at a logarithmic price.

B Reproduction details

The reference implementation (`atlas/`: proper scores; the mixture-betting e-process, the two-sided betting confidence sequence, and the Shiryaev–Roberts e-detector; the horizon-indexed frontier) and the controllable latent simulator are provided, along with all experiment scripts. Synthetic study: `python -m experiments.run_all` regenerates Table 1 and Figures 1, 2. Real data: `python -m experiments.real_data.download` then the `exp_mnist`, `exp_kth`, and `exp_kth.domain` scripts produce Figures 3, 4. GPU neural world models: `python -m experiments.gpu.run_neural --dataset kth --model jepa` (auto-selects CUDA) produces Figure 5; the GPU runs in this paper were executed on a Colab T4. The verified bibliography is in `references.bib`; extended derivations accompany the code in `docs/`.